

9(a)	<i>max 8 of:</i> 1 stick has, pad containing / immobilised, enzymes ; 2 glucose oxidase ; 3 peroxidase ; 4 stick dipped in urine ; A person, urinates / AW, on stick 5 glucose reacts to give hydrogen peroxide ; 6 (hydrogen peroxide reacts with) colourless substance / chromogen ; R dye / pigment 7 to give, colour change / coloured substance ; A change to any named colour 8 compare with colour chart ; 9 more glucose gives darker colour ; 10 specific / only detects glucose ; 11 AVP ; e.g. does not give current <u>blood</u> glucose concentration not numerical	8
------	--	---

Question	Answer	Marks
9(b)	<i>max 7 of:</i> <i>high blood glucose concentration</i> 1 detected by β cells ; I alpha cells I receptors 2 in, islets of Langerhans / pancreas ; 3 (more) insulin secreted ; I produced 4 into blood ; 5 increases glucose absorption in liver (by phosphorylating glucose) ; 6 increases permeability to glucose in, muscle / fat, cells or adds GLUT 4 proteins to cell surface membranes of, muscle / fat, cells ; 7 increases (rate of) respiration of <u>glucose</u> ; 8 conversion of glucose to <u>glycogen</u> / <u>glycogenesis</u> ; 9 inhibits secretion of <u>glucagon</u> / decreases <u>gluconeogenesis</u> ; 10 <u>negative feedback</u> ;	7